

ANISH KUMAR KHARWAR

Senior Software Engineer | Backend & Distributed Systems | GCP · Microservices · Large-Scale Systems

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EXPERIENCE

Senior Software Engineer — Nference | August 2021 – September 2025 | Bengaluru, India

Tech Stack: Python, Bash, Distributed Systems, Pub/Sub, Microservices, GCP, Azure

- **Architected a distributed, queue-based ingestion system** processing **480+ TB** of multi-modal biomedical data, **designing task-isolated pipelines** with independently **scalable publishers and subscribers**, enabling elastic, fault-tolerant horizontal scaling across GCP and Azure.
- **Optimized high-volume image processing workloads (12M+ files)** by integrating image processing models and parallelizing compute-bound tasks across microservices, **improving throughput by ~35%**.
- **Designed a distributed audit and reconciliation framework** to analyze data loss across **45+ TB of structured datasets**, optimizing transformation logic and reducing transformation loss from **23% → 9%**.
- **Extended structured-data microservices** to support **80+ new data categories**, enabling broader ingestion coverage.
- **Improved large-scale document parsing accuracy by 25%+** by redesigning character-order reconstruction algorithms for **700+ GB of PDFs**, applying algorithmic reasoning to handle fragmented text at scale.
- **Owned end-to-end anonymization of new datasets**, collaborating cross-functionally with data providers, domain experts, downstream consumers and approval committees to ensure **regulatory compliance** and **optimal data quality**.
- **Developed 60+ production-grade backend APIs** with RBAC and secure authentication mechanisms, enabling secure multi-tenant access for enterprise users through clear permission hierarchies and access control systems.

Computer Vision & Machine Learning Engineer — Awiros | June 2021 – July 2021 | Gurugram, India

Tech Stack: Python

- **Designed and implemented preprocessing pipelines** for **500K+** images, adding distribution analysis, aspect ratio correction, and data quality validation for ML training.
- **Automated backend workflows** for model training, deployment, and inference, integrating with the UI to streamline the end-to-end ML lifecycle.

Backend Engineering Intern — Vaultedge Software | March 2021 – May 2021 | Bengaluru, India

Tech Stack: Python, Flask, MongoDB, Grafana

- **Migrated job scheduling system** from FIFO to priority-based queues, improving high-priority task latency by **~35%**.
- **Built 6 REST APIs** serving performance metrics to Grafana dashboards and **20+ real-time metrics** to improve system observability.

EDUCATION

Bachelor of Technology, Computer Science and Engineering

Indian Institute of Information Technology, Kalyani (2017 – 2021) | CGPA: 8.35/10

SKILLS

Languages: Python, Bash

Backend: Django, Flask, REST APIs

Architecture: Distributed Systems, Pub/Sub, Microservices, Scalability, Fault Tolerance

Cloud & DevOps: Google Cloud Platform (GCP), Microsoft Azure, Docker, Linux, CI/CD, Git

Databases and Queues: Redis, MySQL, MongoDB, RabbitMQ

Observability: Grafana, Prometheus

INTERESTS

Distributed Computing · Cloud-Native Scalability · High-Throughput, Low-Latency Systems